

Astro 142: Homework 7 - Moffat PSF Fitting

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November, 16, 2025

1 Introduction

This report presents the results of fitting a Moffat point-spread function (PSF) to simulated observational data of a single point source with additive background. The Moffat PSF is given by:

$$\text{PSF}(r) = \frac{\beta - 1}{\pi\alpha^2} \left(1 + \left(\frac{r}{\alpha} \right)^2 \right)^{-\beta} \quad (1)$$

where the model includes five free parameters: the star's position (x_0, y_0) , flux F , scale parameter α , power-law index β , and constant background level B .

2 Methods

A least-squares fitting procedure was implemented using `scipy.optimize.least_squares`. The residual vector was constructed by:

$$\text{residual} = \frac{\text{data} - \text{model}}{\sigma} \quad (2)$$

where σ represents the pixel-wise uncertainty. The 2D coordinate grid was generated using `numpy.meshgrid` and converted to polar coordinates centered on the star position. Multiple initial conditions were tested to assess convergence stability.

3 Results

3.1 Best-Fit Parameters

The least-squares fit converged successfully after 15 function evaluations with a final cost of 8133. Table 1 presents the best-fit parameters with their 1σ uncertainties.

The reduced chi-squared value of $\chi_\nu^2 = 0.497$ indicates a good fit, with the mean normalized residual of 0.0000 and standard deviation of 0.996, consistent with properly estimated uncertainties.

3.2 Fit Stability

Three additional fits were performed with varied initial conditions. All fits converged to the same solution with maximum relative differences $< 0.003\%$. The standard deviations of parameters across different initial conditions were orders of magnitude smaller than the formal uncertainties, demonstrating excellent fit stability. The variation-to-uncertainty ratio was negligible (< 0.01), confirming that the fit is highly insensitive to starting conditions.

Table 1: Best-fit Moffat PSF parameters with 1σ uncertainties.

Parameter	Value	Uncertainty
x_0 (pixels)	54.795	0.081
y_0 (pixels)	45.436	0.081
Flux	7607.6	527.8
α (pixels)	4.702	0.408
β	1.814	0.172
Background	224.079	0.064

3.3 Visual Results

Figure 1 shows the data, best-fit model, and normalized residuals. The model accurately reproduces the observed PSF structure, with residuals distributed symmetrically around zero and no systematic patterns, indicating an appropriate model choice.

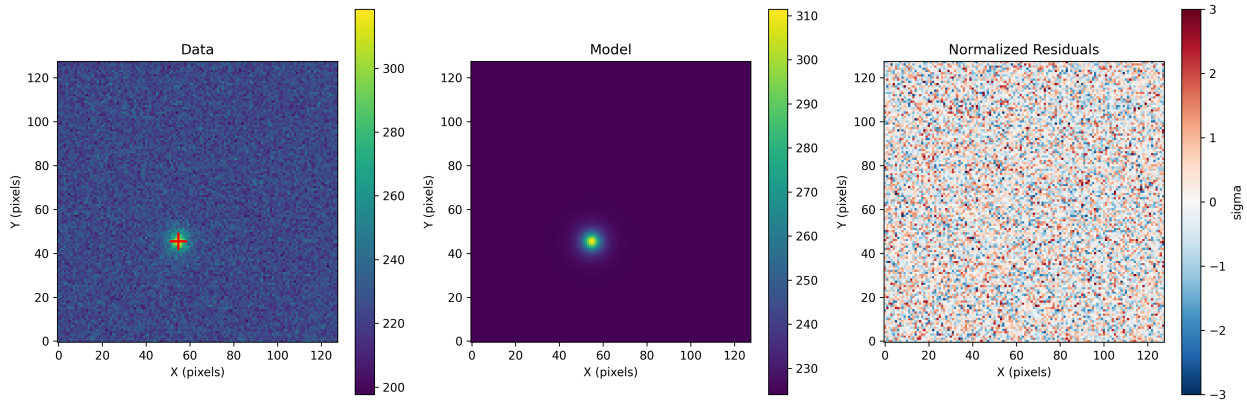


Figure 1: Left: Observed data with fitted star position marked by red cross. Center: Best-fit Moffat PSF model. Right: Normalized residuals showing no systematic structure.

4 Conclusions

The Moffat PSF model successfully describes the point source in the simulated data. The fit is robust, converging reliably from multiple initial conditions to consistent parameter values. The well-behaved residuals and near-unity reduced chi-squared validate both the model choice and uncertainty estimates.